

CLAUDE

Helix OTA — CLAUDE.md

INHERITED FROM constitution/CLAUDE.md

All rules in constitution/CLAUDE.md and the constitution/Constitution.md it references apply unconditionally to this project. Project-specific rules below extend them — they do NOT weaken or override any universal clause.

When this file disagrees with the constitution submodule, the constitution wins.

@constitution/CLAUDE.md

Project overview

Helix OTA is an enterprise-grade over-the-air update system: a custom Go control plane (Gin modular monolith) driving native Android A/B updates (update_engine + AVB/dm-verity + auto-rollback) for RK3588 / Orange Pi 5 Max targets, with a roadmap to Linux, Windows, and other operating systems. Reusable building bricks live in submodules/ (six ota-* modules) and the dev/runtime infrastructure in containers/.

Project-specific MANDATORY constraints

Build / packaging

- Server: Go module `github.com/HelixDevelopment/helix_ota/server` (server/). `go build ./...`, `gofmt`, `go vet` MUST be clean.
- Submodule Go bricks build + test from their own module roots under `submodules/`.
- Android bricks (ota-update-engine-bridge, ota-android-agent) build with AGP 8.5.2 + plain `kotlin.android` on Gradle 9.5 (Kotlin MPP does NOT build on Gradle 9.5 — see CONTINUATION handoff).
- Documentation source of truth is Markdown + Mermaid; PDF/HTML/DOCX and draw.io/SVG/UML/PNG are generated artifacts (`scripts/export_docs.sh`).

Test / verification

- Server: `cd server && go test ./...` (httptest integration suite).
- Inheritance gate (this wiring): `bash tests/inheritance_gate.sh`.
- Full inheritance + paired-mutation proof: `bash tests/test_constitution_inheritance.sh`.

Deployment / flash / publish

- Multi-upstream: the parent and every owned submodule push to all four upstreams (GitHub primary + GitLab + GitFlic + GitVerse). The parent's origin remote fans out pushes to all four.

Project-specific architecture notes

- Persistence seam: `server/internal/store.Repository`. MVP wires the in-memory implementation; the `pgx/PostgreSQL` implementation is the production target (architecture.md §4).
 - Trust boundary (security): the artifact-signature verification key comes ONLY from server configuration — never from the request (`server/internal/api/handlers_artifact.go:resolvePublicKey`).
-

Project overrides of universal rules

(none — this project does not override any universal clause)